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Heat Exchanger

Objectives

- Create a SolidWorks model of a double-pipe heat exchanger using given dimensions
- Simulate the heat exchangers of co-current and counter-current orientations
- Plot the non-dimensional velocity profiles and laminar theory of the pipe flows and the annular flow separately
- Determine the total rate of heat transfer using the bulk enthalpies obtained from simulations
- Compare the rate of heat transfer of the hot flow and the cold flow
- Calculate the overall heat transfer to determine the rate of heat for both co-current and counter-current representations
- Compare the rate of heat transfer found by using SolidWorks simulation and analytical calculation
- Utilize the analysis to discuss which flow configuration transfers the most heat and which method is more accurate

Background

- ❖ Steady-state macroscopic energy balance

$$\Delta\left(\hat{H} + \frac{1}{2}v^2 + gh\right) = \hat{Q} + \hat{W}_m$$

Where:

$\hat{H}$: specific enthalpy (J/kg)

v: velocity of fluid (m/s)

g: gravity (m/s²)

h: height (m)

$\hat{Q}$: heat per unit mass (J/kg)

$\hat{W}_m$: work is done per unit mass (J/kg)

Since the system is set up as a steady-state using water (incompressible) for the fluids, the change in kinetic energy is negligible. The change in potential energy is zero as the system (horizontal flows) does not change in elevation. There is no mechanical work being done on the system; therefore, $\hat{W}_m = 0$. Thus,

$$\Delta\hat{H} = \hat{Q}$$

$$w(\Delta\hat{H}) = w(\hat{Q})$$

$$\dot{H}_{in} - \dot{H}_{out} = Q$$

Where:

Q: heat loss or heat gain of the fluid (W)

$\dot{H}$: rate of enthalpy (W)

w: mass flow rate (kg/s)

❖ Determine the Reynolds number of the pipe flow

$$Re_i = \frac{\rho v D}{\mu}$$

Where:

Re: Reynolds number

ρ : density of the fluid (kg/m³)

D: diameter of the pipe

μ : dynamic viscosity of the fluid (Pa · s)

❖ Determine the Reynolds number of the annular flow

Since the annular flow does not have a specific derived equation, the mean hydraulic radius is used to determine the Reynolds number. This method is often used for turbulent flow in a non-circular tube. Thus, $R_h = \frac{S}{Z}$

Where:

R_h : mean hydraulic radius (m)

S: cross-sectional area of the annulus (m²)

Z: wetted perimeter (m)

$$Re_h = \frac{4R_h v \rho}{\mu}$$

After finding the mean hydraulic radius, the Reynolds number of the outer flow can be found using the equation above.

❖ Determine the overall heat transfer coefficient of the flow in the pipe

Reynolds and Prandtl numbers are needed to determine the convection heat transfer coefficient. Thus,

$$Pr = \frac{\hat{C}_p \mu}{k}$$

Where:

Pr: Prandtl number

$\hat{C}_p$: specific heat at constant pressure (J/kg · K)

k: thermal conductivity (W/m · K)

$$\frac{h_{in} D}{k} (Re)^{-1} (Pr)^{-\frac{1}{3}} \left(\frac{\mu_b}{\mu_0} \right)^{-0.14} = C$$

Where:

h_{in} : convective heat transfer coefficient (W/m² · K)

D: outer diameter of the pipe

C: constant which can be found from figure 14.3-2

$$T_{ln} = \frac{(T_{h2} - T_{c2}) - (T_{h1} - T_{c1})}{\ln \ln \left[\frac{T_{h2} - T_{c2}}{T_{h1} - T_{c1}} \right]}$$

Where:

ΔT_{ln} : temperature difference (K)

T_{h2} : temperature of the hot fluid at plane 2 (K)

T_{c2} : temperature of the cold fluid at plane 2 (K)

T_{h1} : temperature of the hot fluid at plane 1 (K)

T_{c1} : temperature of the cold fluid at plane 1 (K)

It is important to note that the ΔT_{ln} of the counter-current orientation is different from that of the co-current representation.

$$\frac{1}{D_0 U_0} = \frac{1}{D_0 h_0} + \frac{\ln \ln \left(\frac{D_1}{D_0} \right)}{2k_{01}} + \frac{1}{D_1 h_1}$$

Where:

U_0 : overall heat transfer coefficient ($W/m^2 \cdot K$)

D_0 : inner diameter of the pipe (m)

h_0 : convective heat transfer coefficient of the pipe flow ($W/m^2 \cdot K$)

D_1 : outer diameter of the pipe (m)

h_1 : convective heat transfer coefficient of the annular flow ($W/m^2 \cdot K$)

k_{01} : thermal conductivity of Stainless Steel 302 ($W/m \cdot K$)

With the overall heat transfer coefficient and the temperature difference, the heat transferred can be found by using the equation below.

$$Q_c = -Q_h = U_0 (2\pi R_0 L) (T_{ln})$$

Results and Analysis

❖ Velocity Profiles

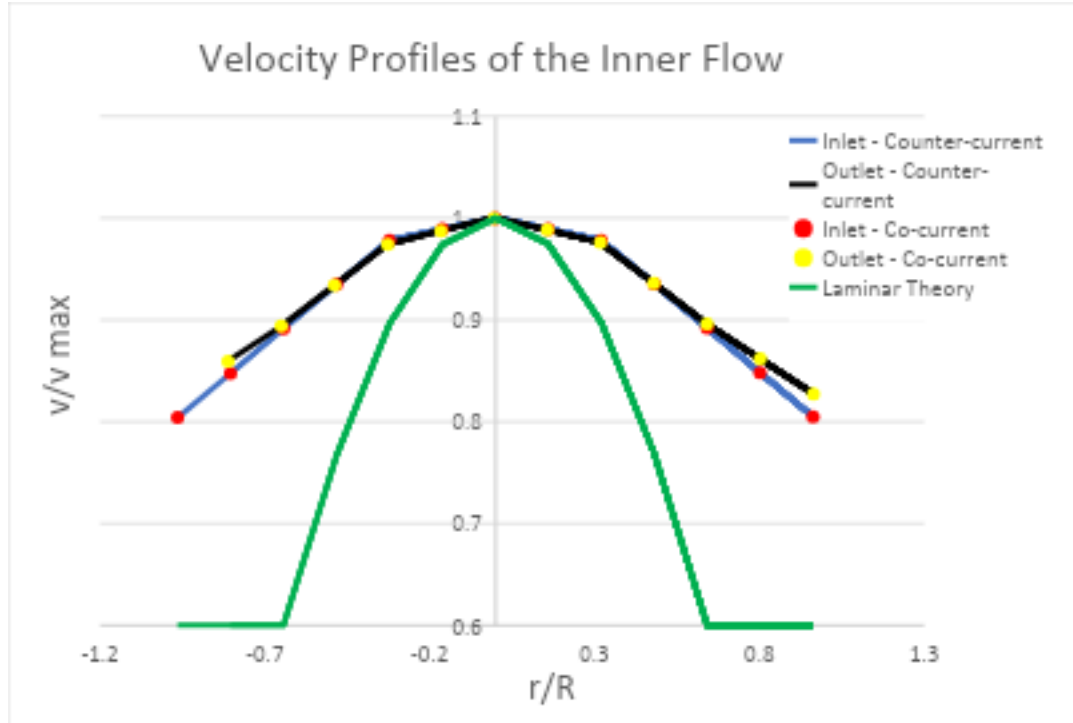


Figure 1: Velocity Profiles of the Inner Flow of Counter-current and Co-current Representations

The first figure shows the velocity profiles of both representations and the laminar theory of the flow in a pipe. The co-current flows are presented as sets of dots while the counter-current flows are presented as two solid lines. This is because the lines overlap one another and it would be difficult to analyze if all lines were solid. The outlet of co-current data (yellow dots) traits along with the outlet of counter-current line (black) precisely. Similarly, the inlet of co-current data (red dots) overlays on the inlet of the counter-current line (blue).

Generally, the velocity profiles of all four flows are very similar to one another. This confirms that the simulation was set up correctly based on the assumption of constant velocity. Though temperature does change among the pipe, the fluid density does not vary significantly. Thus, the velocity is constant in this steady system. Besides, all four flows are turbulent because none of them is close to the bell curve of the laminar theory line (green). The flows confirm that the fluid velocity is highest in the middle and lowest at the wall of the pipe.

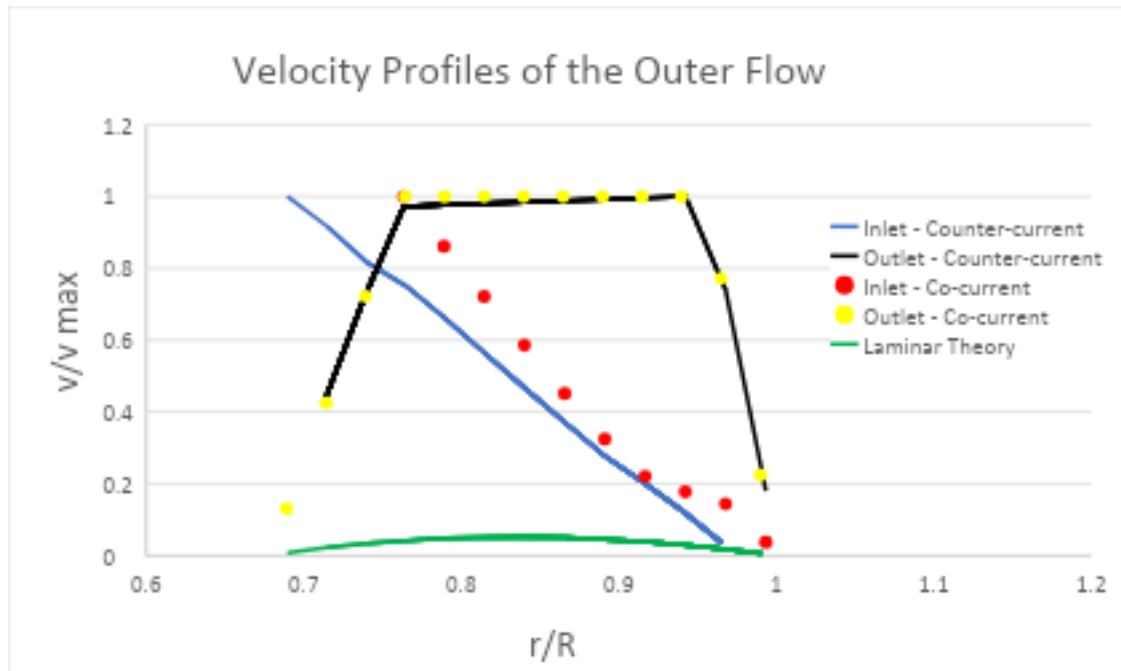


Figure 2: Velocity Profiles of the Outer Flow of Counter-current and Co-current Representations

Figure 2 presents the velocity profiles of both represents and a laminar theory curve (green) of the flow in an annular region. The co-current flows are presented as sets of dots while the counter-current flows are introduced as two solid lines. This set provides a clearer presentation for detail analysis. Due to the size of the collected data, the laminar theory is not shown a clear bell curve in the desired region. All four flows are turbulent as they are very different from the laminar theory. The outlet velocity of both co-current and counter-current are very close to each other while the inlet velocity profiles show a small difference. This difference is often caused by the numerical noises of SolidWorks simulation.

The inlet line and dots demonstrate that the inlet velocity decreases as the distance from the pipe increases. Theoretically, fluid velocity should be the same as the fluid has just entered the pipe meaning that the inlet lines should have straight across the graph. The biggest difference between the highest and lowest inlet velocities of counter-current and co-current are 3 m/s and 1.5 m/s respectively. These values are significant enough to cause errors in the data gathered for later analysis. The outlet line and dots of fully developed flows show that the velocities are similar across the annular and smaller as the fluid is closer to the wall.

❖ Temperature Profiles

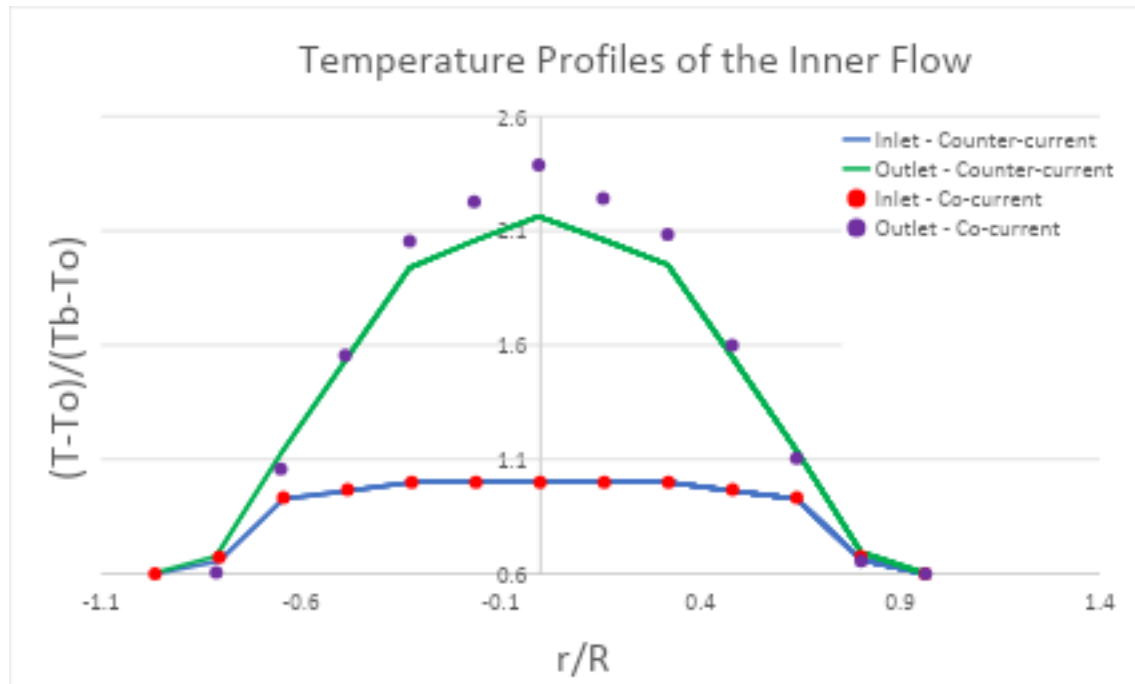


Figure 3: Temperature Profiles of the Inner Flow of Counter-current and Co-current Representations

The figure above presents the temperature profiles of counter-current and co-current representations of the flow in a pipe. With the dots being co-current and lines being counter-current flows. Both inlet profiles demonstrate the same prediction as the red dots overlap the blue line. The data indicates that the temperature is expected to be the same across the pipe as the fluid has just entered. At the outlet, the temperature is highest in the middle and lowest along the pipe wall. This follows the idea that heat is being transferred from the pipe to the annulus causing the temperature to decrease when moving from the mid-point to the wall.

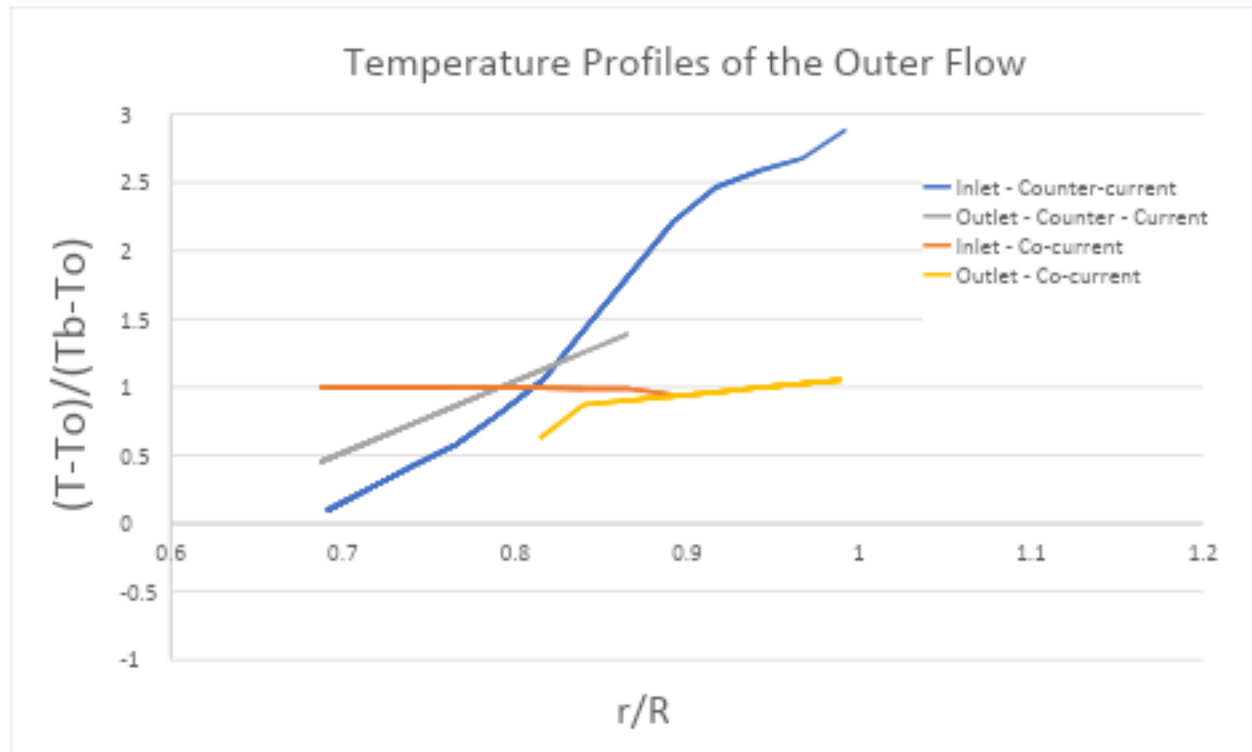


Figure 4: Temperature Profiles of the Outer Flow of Counter-current and Co-current Representations

Figure 4 shows the temperature profiles of both representations of the annular flow. This graph is the hardest graph to read as the obtained data is not very accurate and precise. The inlet of the counter-current line (blue) shows an upward slope when probing from the inner wall (pipe's wall) to the outer wall (the inner annulus' wall). At the inlet, the temperature across the annular area should be the same meaning that the blue line should have been a straight line cross. Though the blue line is upward, the difference between its highest and lowest temperatures is only 0.07% which is very negligible. Again, this could have been caused by SolidWorks simulation.

The outlet of the counter-current representation also gives an upward slope; however, it is accurate this time. The grey line is a bit short as some data were too close to the wall confusing the shape of the graph. It is expected that the temperature decreases as the distance from the pipe's wall become larger. This trend is shown from the collected data table; however, the line presents an upward slope as the bulk temperature is smaller than the wall temperature.

The inlet of the co-current orientation is more accurate than the counter-current representation. The orange line is straight across the graph as expected. The outlet of co-current flow (yellow) also demonstrates a similar trend as the outlet of the counter-current flow (grey). It

is noteworthy that the slope of the counter-current flow (grey) is steeper than the slope of the co-current flow (yellow). This explains that the heat transferred is more efficient in the counter-current orientation. However, the co-current simulation provides accurate and consistent data while the counter-current simulation includes a lot of numerical noises.

❖ Rate of Heat Transfer

Table 1: Heat Gain and Heat Loss of Counter-current and Co-current Orientations

	Counter-current	Co-current
Q_c (W)	1893	2228
Q_h (W)	-2229	-2183

It is important to note that minus indicates a heat loss, specifically heat is being transferred from the pipe to the annulus. As indicated in the background section, $Q_c = -Q_h$ meaning that the counter-current simulation has a lot of numerical noises because its heat gain and heat loss have a significant gap between them (about -336W). Meanwhile, the co-current simulation provides a more accurate and precise data as the two values are very close to each other. During the analysis from the four graphs above, it was noted that the data obtained from the pipe is more consistent and accurate than that of the annulus. Therefore, Q_h values of counter-current and co-current are used for further analysis in the comparison section below.

❖ Overall Heat Transfer Coefficient

Calculations for the inner flow in the pipe

$$v_i = \frac{Q}{A} = \frac{Q}{\frac{\pi}{4} D_1^2} = \frac{3.4 \frac{L}{min} \times \frac{1min}{60s} \times \frac{1m^3}{(10dm)^3}}{\frac{\pi}{4} (0.0083m)^2} = 1.047 \frac{m}{s}$$

$$Re_i = \frac{\rho v D}{\mu} = \frac{965.81 \frac{kg}{m^3} \times 1.047 \frac{m}{s} \times 0.0083m}{0.0003 Pa \cdot s} = 27978 \approx 2.8 \times 10^4$$

$$Pr = \frac{\hat{C}_p \mu}{k} = \frac{4182 \frac{J}{kg \cdot K} \times 0.0003 Pa \cdot s}{672.88 \times 10^{-3} \frac{W}{mK}} = 1.865 \quad [1] [2]$$

From the figure 14.3-2

$$\frac{h_{in} D}{k} (Re)^{-1} (Pr)^{-\frac{1}{3}} \left(\frac{\mu_b}{\mu_0} \right)^{-0.14} = 0.0034$$

$$h_{ln} = \frac{0.0034 k}{D_1 (Re_i)^{-1} (Pr)^{-\frac{1}{3}} \left(\frac{\mu_b}{\mu_0}\right)^{-0.14}} = \frac{0.0034 \times 672.88 \times 10^{-3} \frac{W}{mK}}{0.0083m \times (27978)^{-1} (1.865)^{-\frac{1}{3}} (1)^{-0.14}} = 9493 \frac{W}{m^2 K}$$

$$h_0 = h_{ln} = 9493 \frac{W}{m^2 K}$$

Calculations for the outer flow in the annular

$$v_o = \frac{Q}{A} = \frac{Q}{\frac{\pi}{4} (D_3 - D_2)^2} = \frac{3.586 \frac{L}{min} \times \frac{1min}{60s} \times \frac{1m^3}{(10dm)^3}}{\frac{\pi}{4} (0.014^2 - 0.0095^2)} = 0.7196 \frac{m}{s}$$

$$R_h = \frac{S}{Z} = \frac{\frac{\pi}{4} (D_3 - D_2)^2}{2\pi (R_3 + R_2)} = \frac{\frac{\pi}{4} (0.014m^2 - 0.0095m^2)}{2\pi \left(\frac{0.014m}{2} + \frac{0.0095m}{2}\right)} = 1.125 \times 10^{-3} m$$

$$Re_o = Re_h = \frac{4R_h v_p}{\mu} = \frac{4 \times 1.125 \times 10^{-3} m \times 0.7196 \frac{m}{s} \times 998.69 \frac{kg}{m^3}}{0.0011 Pa s} = 2940 \approx 2.9 \times 10^3$$

$$Pr = \frac{\hat{C}_p \mu}{k} = \frac{4182 \frac{J}{kg K} \times 0.0011 Pa s}{588.34 \times 10^{-3} \frac{W}{mK}} = 7.82 \quad [1]$$

From figure 14.3-2

$$\frac{L}{D_2} = \frac{1.32m}{0.0095m} \approx 140$$

$$\frac{h_{ln} D}{k} (Re)^{-1} (Pr)^{-\frac{1}{3}} \left(\frac{\mu_b}{\mu_0}\right)^{-0.14} = 0.003$$

$$h_{ln} = \frac{0.003 k}{D_2 (Re_i)^{-1} (Pr)^{-\frac{1}{3}} \left(\frac{\mu_b}{\mu_0}\right)^{-0.14}} = \frac{0.003 \times 588.34 \times 10^{-3} \frac{W}{mK}}{0.0095m \times (2940)^{-1} (7.82)^{-\frac{1}{3}} (1)^{-0.14}} = 1084 \frac{W}{m^2 K}$$

$$h_1 = h_{ln} = 1084 \frac{W}{m^2 K}$$

$$k_{01} = 16.2 \frac{W}{m K} \quad [3]$$

$$\frac{1}{D_0 U_0} = \frac{1}{D_0 h_0} + \frac{\ln \ln \left(\frac{D_1}{D_0}\right)}{2k_{01}} + \frac{1}{D_1 h_1} = \frac{1}{0.0083m \times 9493 \frac{W}{m^2 K}} + \frac{\ln \ln \left(\frac{0.0095m}{0.0083m}\right)}{216.2 \frac{W}{m K}} + \frac{1}{0.0095m \times 1084 \frac{W}{m^2 K}}$$

$$\frac{1}{D_0 U_0} = 0.11397 \frac{mK}{W}$$

$$U_0 = \frac{1}{0.0083m \times 0.11397 \frac{mK}{W}} = 1057 \frac{W}{m^2 K}$$

Heat gain/loss of counter-current flow

$$T_{ln} = \frac{(T_{h2}-T_{c2})-(T_{h1}-T_{c1})}{\ln \ln \left[\frac{T_{h2}-T_{c2}}{T_{h1}-T_{c1}} \right]} = \frac{(353.43-288.15)-(363.15-295.66)}{\ln \ln \left[\frac{353.43-288.15}{363.15-295.66} \right]} = 66.38K$$

$$Q_c = -Q_h = U_0(2\pi R_0 L)(T_{ln}) = 1057 \frac{W}{m^2 K} \times \left(2\pi \frac{0.0083m}{2} \times 1.32m \right) (66.38K) = 2415 W$$

Heat gain/loss of co-current flow

$$T_{ln} = \frac{(T_{h2}-T_{c2})-(T_{h1}-T_{c1})}{\ln \ln \left[\frac{T_{h2}-T_{c2}}{T_{h1}-T_{c1}} \right]} = \frac{(353.66-297.07)-(363.15-288.15)}{\ln \ln \left[\frac{353.66-297.07}{363.15-288.15} \right]} = 65.36K$$

$$Q_c = -Q_h = U_0(2\pi R_0 L)(T_{ln}) = 1057 \frac{W}{m^2 K} \times \left(2\pi \frac{0.0083m}{2} \times 1.32m \right) (65.36K) = 2378 W$$

❖ Comparison of the Qs

Table 2: Heat transferred of two orientations of two methods and percent errors

	Counter-current	Co-current
Solidworks Q (W)	2229	2183
Analytical Q (W)	2415	2378
Percent Error (%)	7.7	8.2

Both analytical method and SolidWorks simulation show that the counter-current orientation transfers more heat than the co-current configuration, specifically 37 W and 46 W respectively.

❖ Comparison of two methods

The heat loss/gain obtained by two methods provide similar values. The average percent error is about 8% for both configurations (see table 2). The simulation was helpful when finding the fluid properties (e.g. density, viscosity, etc.) simply by probing the fluid at a specific location. It is reasonable to say that both methods are accurate; however, the analytical method is preferred as it is more precise. In addition, SolidWorks simulation has shown a lot of numerical noises causing confusion when analyzing the data. In other words, the hand-calculation method is more accurate in this analysis.

Conclusion

In conclusion, indirect heating/cooling was demonstrated by a double-pipe heat exchanger SolidWorks model. The velocity and temperature profiles of SolidWorks simulation meet the theoretical expectation. The rate of heat transfer was found by using both SolidWorks

and hand-calculation. The SolidWorks simulation showed some inaccurate data, specifically in the annular flow due to numerical noises. Meanwhile, the hand-calculation method gives a more accurate result. It is also easier to keep track of the process by using the analytical method. Thus, hand-calculation is preferred in this analysis. The results of both methods also confirm that the counter-current configuration transfers more heat compared to the co-current orientation. This result follows the principle provided by the Introductory Transport Phenomena book [4]. For future applications, a counter-current representation is a great start when designing a double-pipe heat exchanger. Each method has certain advantages; therefore, both techniques should be used to compare and ensure the accuracy of the obtained data.

References

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